TEST CASE DESIGN
& EXECUTION REPORT

Online Book Store Project

Assignment No. 2

Submitted By	Farooque Sajjad
Department	Quality Assurance
Organization	10Pearls
Reference	Assignment 1 - Requirement Analysis
Test Tool	Testworthy (Test Management) And Excel
Submission Date	05-01-2026

1. Document Control

This document covers test case design and execution for the Online Book Store project. It is a continuation of Assignment 1, which delivered the requirement analysis and user stories. All test cases in this report are linked to the functional requirements and user stories defined in that submission.

Version	Date	Prepared By	Reviewed By	Status
1.0	05-01-2026	Farooque Sajjad	[Supervisor Name]	Draft

2. Project Summary

The Online Book Store is a web-based e-commerce platform that allows users to browse, search, and purchase books. The system supports three distinct user roles: Guest, Registered Customer, and Administrator. Assignment 1 produced 24 functional requirements, 12 non-functional requirements, 24 user stories, and a requirements traceability matrix across four planned sprints.

This assignment builds on that foundation by designing test cases for all identified functional requirements, setting up a test management environment in Testworthy & writing testcases in Excel , organizing tests into suites and runs aligned with the sprint plan, and recording execution results.

2.1 Scope of Testing

- User authentication including registration, login, and password reset
- Book catalog browsing, search, filter, and sort for both guests and registered users
- Shopping cart operations and the full checkout flow
- Payment processing including card and mobile wallet scenarios
- Order confirmation, order history, and customer notifications
- Wishlist and book review functionality
- All three administrator modules: catalog, order, and user management
- Extended features including discount codes and book recommendations

2.2 Out of Scope

- Automated test execution (no automation framework was set up for this phase)
- Performance and load testing
- Mobile application testing (FR-21 is a Won't Have item)
- Security penetration testing

3. Test Methodology

The testing approach for this assignment followed standard QA practices used in agile software development. The methodology is described below.

3.1 Testing Approach

Black-box functional testing was the primary approach used throughout. Test cases were designed based on the acceptance criteria written in Assignment 1 without any knowledge of the underlying implementation. Each test case maps directly to a user story and its linked functional requirement.

The following testing types were applied across the test suite:

Test Type	Purpose	Applied To
Smoke	Verify core flows work after a build	Login, Registration, Catalog load, Payment success, Admin catalog
Functional	Verify each feature works per stated requirements	All 67 test cases cover functional verification
Negative	Verify system handles invalid input and errors correctly	Duplicate email, wrong password, empty cart checkout, expired card
Regression	Verify existing features not broken by new changes	Cart persistence after logout, re-login flow
Boundary	Verify system behaviour at edge of valid ranges	Password length, price range filter, 5 failed login attempts

3.2 Test Design Technique

Equivalence Partitioning and Boundary Value Analysis were the primary test design techniques used. For each input field, valid and invalid partitions were identified and at least one test case was written per partition. For sequential flows such as checkout and payment, end-to-end scenario testing was applied.

Acceptance criteria from the Gherkin scenarios in Assignment 1 were used directly as the basis for writing expected results. This ensured that every test case has a clear pass or fail condition that maps back to a documented requirement.

3.3 Test ID Convention

Test cases follow a structured naming convention to make them easily traceable:

Format	Example and Meaning

TC-[MODULE]-[NNN]	TC-AUTH-001: Test Case, Authentication module, sequence 001
Modules Used	AUTH, CAT, CART, PAY, WR, ADM, EXT

3.4 Priority Classification

Each test case was assigned a priority based on the business impact of the feature it covers:

Priority	Criteria
High	Feature is a Must Have requirement. Failure blocks core user journeys or transactions.
Medium	Feature is a Should Have requirement. Failure impacts usability but does not block core flows.
Low	Feature is a Could Have requirement. Failure has minimal user impact in the current release.

4. Test Environment and Tool Setup

4.1 Test Management Tool: Excel

The test cases were first authored in Microsoft Excel before being entered into Testworthy. The Excel file serves as the primary offline reference document and contains all 67 test cases organized across seven module-specific sheets. Each sheet corresponds to a test suite in Testworthy.

The Excel document follows the same column structure used in industry-standard manual test case templates. Each row represents one test case with the following fields: Test Case ID, Linked User Story, Linked Functional Requirement, Test Objective, Test Type, Priority, Precondition, Test Steps, Test Data, Expected Result, Actual Result, and Status.

A dedicated Document Info sheet at the start of the file contains the product name, reference document, prepared by, date of creation, and a legend explaining the status codes and test type classifications used throughout the file. An Execution Summary sheet at the end of the file automatically calculates the pass count and pass rate per module using Excel formulas.

The Excel file was maintained in parallel with Testworthy throughout the assignment. Since Testworthy does not support test case import in the free tier, the Excel document acted as the authoring environment where test cases were drafted, reviewed, and refined before being manually entered into the tool. This approach ensured that a structured and formatted offline backup was available at all times.

4.2 Test Management Tool: Testworthy

Testworthy was selected as the test management platform for this assignment. It was set up on a 30-day free trial using an institutional email account. All 67 test cases were manually entered into the platform and organized into suites and runs aligned with the sprint plan from Assignment 1.

Note on import/export: Testworthy does not provide a CSV or Excel import/export option for test cases in the free trial tier. As a result, all 67 test cases were entered manually into the tool. While time-consuming, this process provided a thorough understanding of the tool's structure including suites, sections, runs, milestones, and test plans.

4.2.1 Project Configuration in Testworthy

Setting	Value
Project Name	Online Book Store Phase 01
Description	End-to-end functional testing covering User Auth, Catalog, Checkout, and Admin modules
Total Test Cases	67
Total Suites	7 (AUTH, CAT, CART, PAY, WR, ADM, EXT)
Total Milestones	4 (one per sprint)
Test Plans	1 (Master Functional Test Plan)
Test Runs	5 (per sprint phase)
Test Type Options Available	Functional, Regression, Performance, Security, Automation

Note on test type classification: Testworthy's free tier only provides five test type options in its dropdown. Types such as Negative, Smoke, Sanity, and Boundary are not available as distinct options. To work within this constraint, all test cases including negative and smoke scenarios were classified as Functional in Testworthy. The actual test type for each case is documented in the Excel test case file and in this report.

4.2.2 Testworthy Project Screenshot

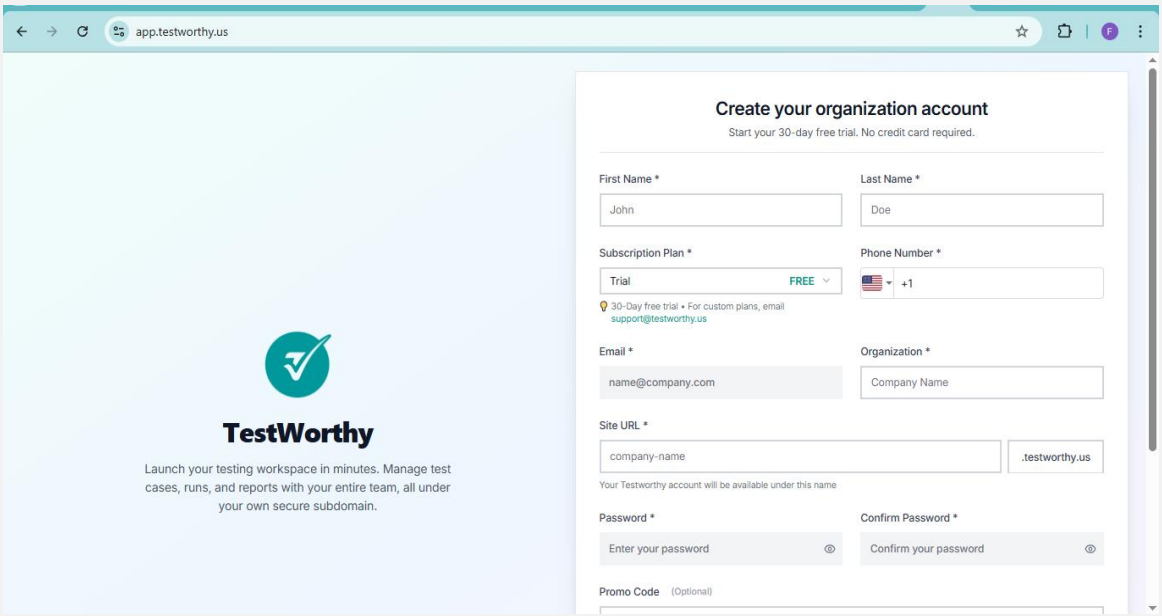


Figure 1: Creating an account onTestworthy

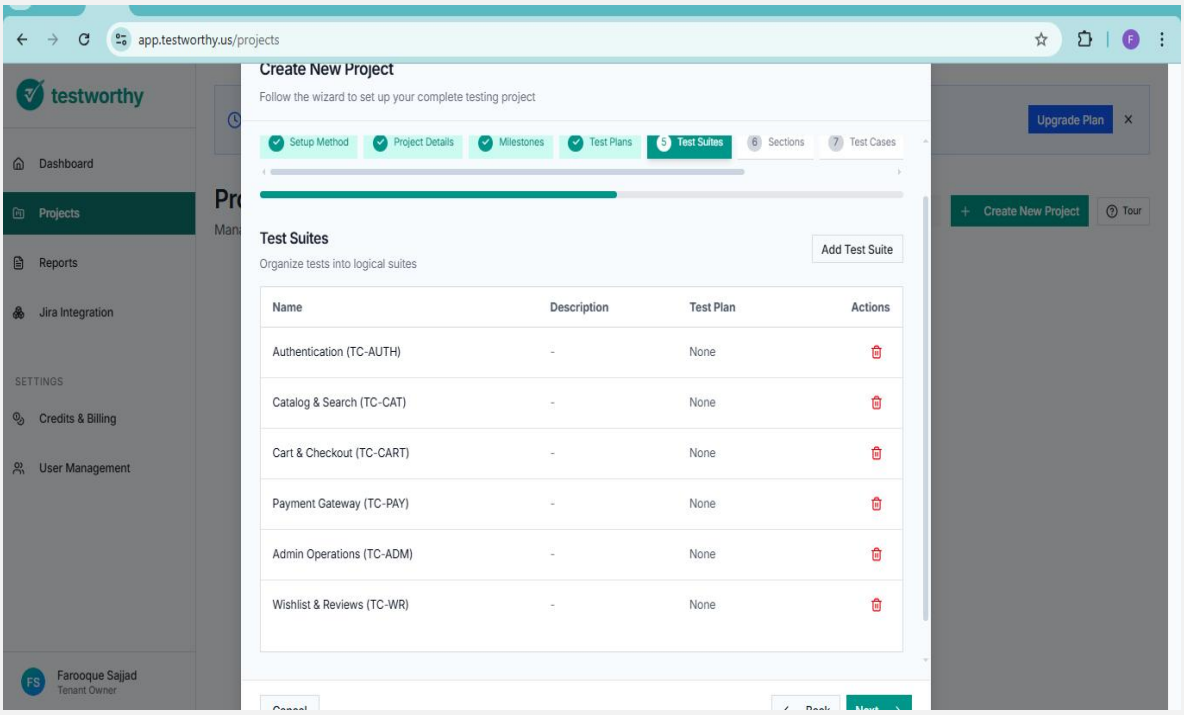


Figure 2: Testworthy project up

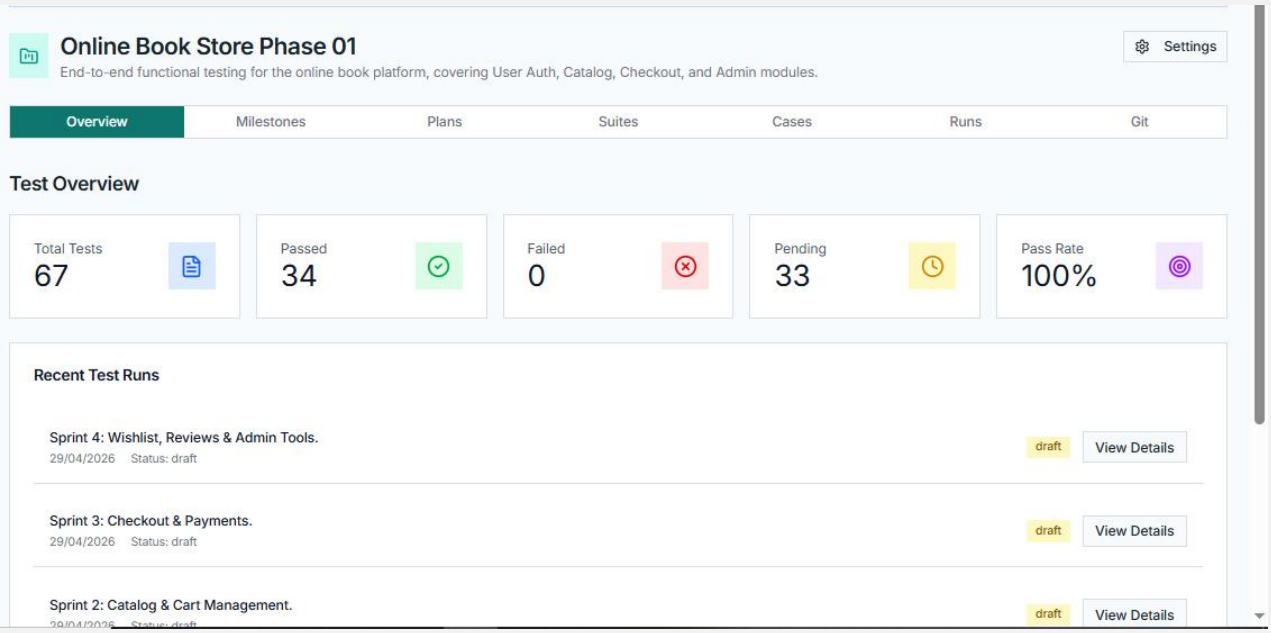


Figure 3: Testworthy project overview showing 67 total test cases, 34 passed, 33 pending, and 100% pass rate

5. Test Suites and Case Distrubution

Test cases were organized into seven suites in Testworthy, each corresponding to a functional module of the Online Book Store. The table below summarizes the distribution.

Suite ID	Suite Name	Test Cases	Linked FRs	Sprints
TC-AUTH	Authentication	11	FR-01 to FR-03	Sprint 1
TC-CAT	Catalog & Search	13	FR-04 to FR-07	Sprint 1 & 2
TC-CART	Cart & Checkout	8	FR-08 to FR-09	Sprint 2 & 3
TC-PAY	Payment & Orders	9	FR-10 to FR-12	Sprint 3
TC-WR	Wishlist & Reviews	7	FR-13 to FR-14	Sprint 4
TC-ADM	Admin Functions	11	FR-15 to FR-17	Sprint 2 to 4
TC-EXT	Notifications, Discounts & Recommendations	8	FR-18 to FR-20	Sprint 3 & 4
	Total	**67**		

5.1 Test Suites Screenshot

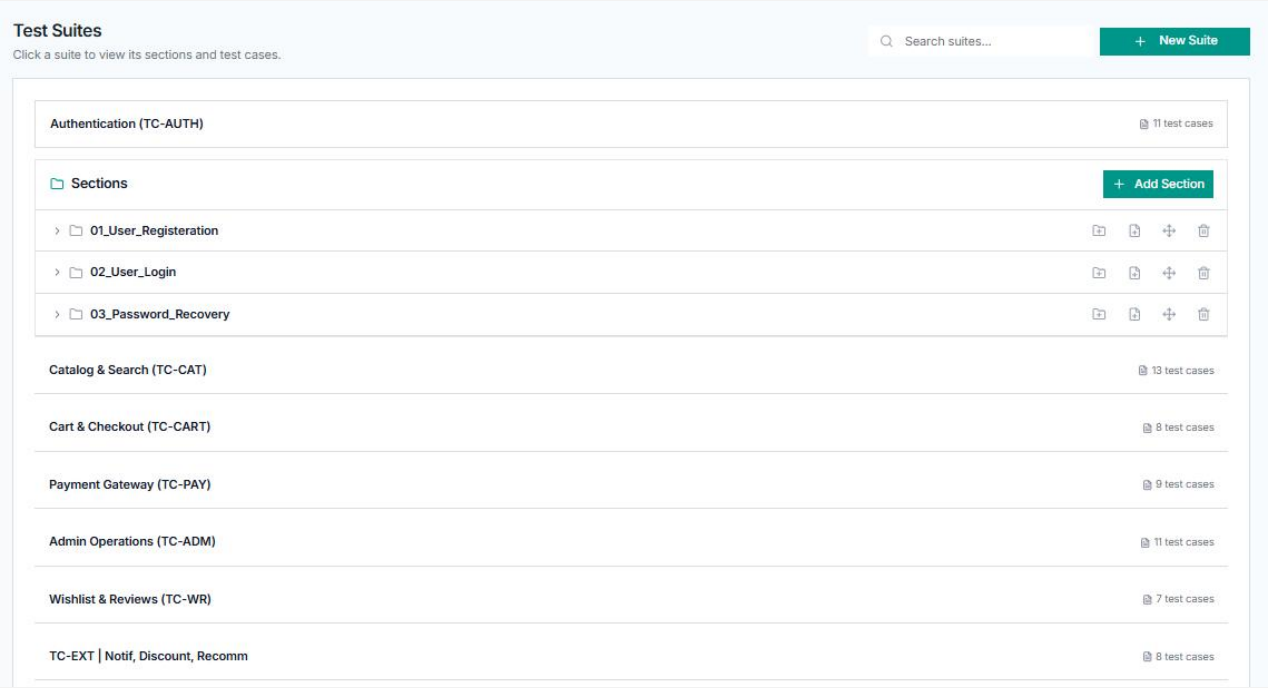


Figure 4: Testworthy suites view showing all 7 suites with case counts

6. Test Case Summary

The full set of 67 test cases was designed using the following structure for each entry: Test Case ID, Linked User Story, Linked Functional Requirement, Test Objective, Test Type, Priority, Precondition, Test Steps, Test Data, Expected Result, Actual Result, and Status. The complete test case document is maintained in the accompanying Excel file.

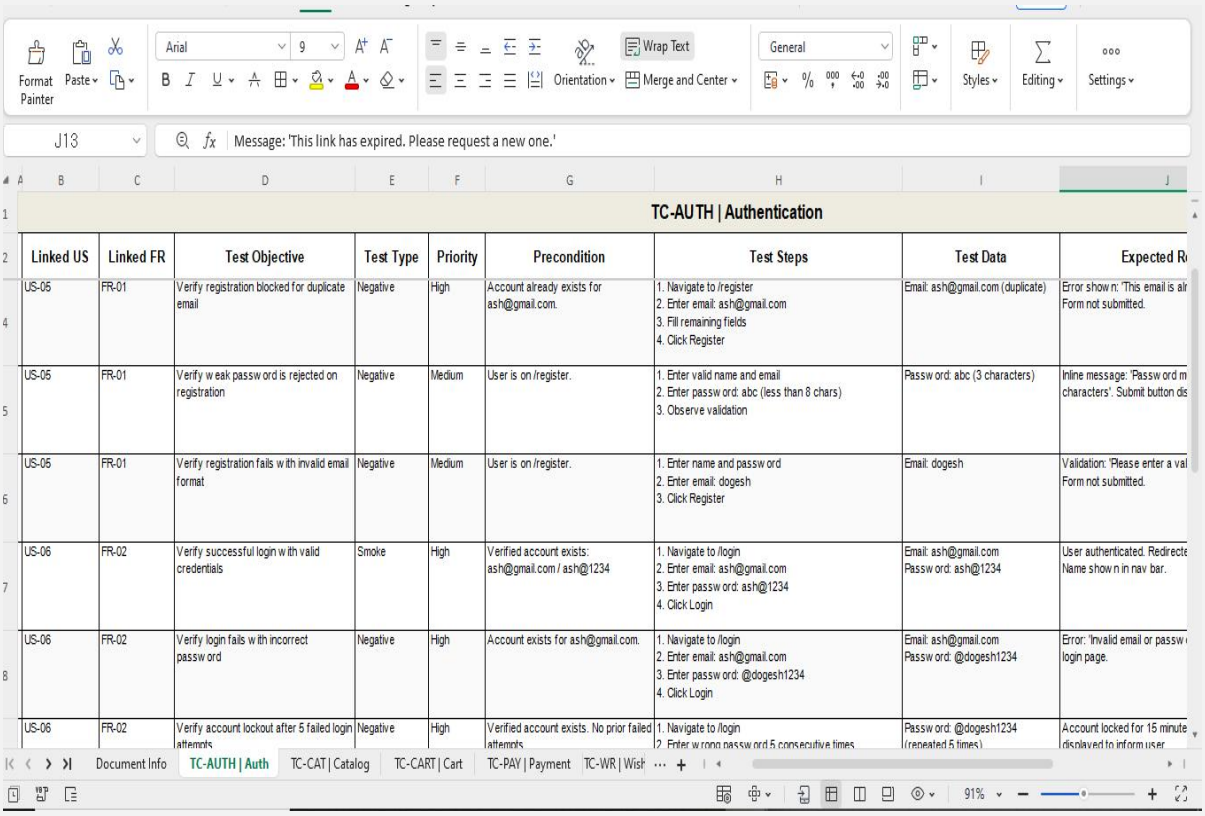
Below is a representative sample of test cases from each module to illustrate the format and coverage.

6.1 Sample Test Cases

Authentication Module (TC-AUTH)

TC ID	US	FR	Objective	Type	Status
TC-AUTH-001	US-05	FR-01	Verify successful registration with valid data	Smoke	**Pass**
TC-AUTH-002	US-05	FR-01	Verify registration blocked for duplicate email	Negative	**Pass**
TC-AUTH-005	US-06	FR-02	Verify successful login with valid credentials	Smoke	**Pass**

TC-AUTH-007	US-06	FR-02	Verify account lockout after 5 failed login attempts	Negative	**Pass**
TC-AUTH-011	US-07	FR-03	Verify expired reset link is rejected	Negative	**Pass**



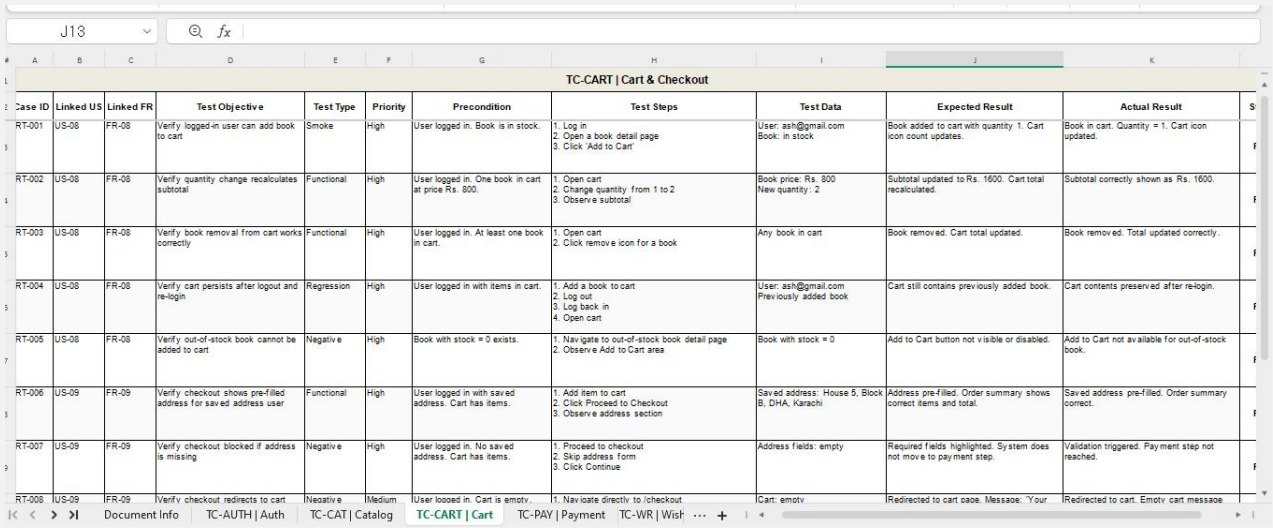
TC-AUTH | Authentication

Linked US	Linked FR	Test Objective	Test Type	Priority	Precondition	Test Steps	Test Data	Expected R
US-05	FR-01	Verify registration blocked for duplicate email	Negative	High	Account already exists for ash@gmail.com.	1. Navigate to /register 2. Enter email: ash@gmail.com 3. Fill remaining fields 4. Click Register	Email: ash@gmail.com (duplicate)	Error show n: 'This email is al Form not submitted.
US-05	FR-01	Verify w eak passw ord is rejected on registration	Negative	Medium	User is on /register.	1. Enter valid name and email 2. Enter passw ord: abc (less than 8 chars) 3. Observe validation	Passw ord: abc (3 characters)	Inline message: 'Passw ord m characters'. Submit button dis
US-05	FR-01	Verify registration fails w ith invalid email format	Negative	Medium	User is on /register.	1. Enter name and passw ord 2. Enter email: dogesh 3. Click Register	Email: dogesh	Validation: 'Please enter a val Form not submitted.
US-06	FR-02	Verify successful login w ith valid credentials	Smoke	High	Verified account exists: ash@gmail.com / ash@1234	1. Navigate to /login 2. Enter email: ash@gmail.com 3. Enter passw ord: ash@1234 4. Click Login	Email: ash@gmail.com Passw ord: ash@1234	User authenticated. Redirecte Name show n in nav bar.
US-06	FR-02	Verify login fails w ith incorrect passw ord	Negative	High	Account exists for ash@gmail.com.	1. Navigate to /login 2. Enter email: ash@gmail.com 3. Enter passw ord: @dogesh1234 4. Click Login	Email: ash@gmail.com Passw ord: @dogesh1234	Error: 'Invalid email or passw login page.
US-06	FR-02	Verify account lockout after 5 failed login attempts	Negative	High	Verified account exists. No prior failed attempts	1. Navigate to /login 2. Enter w rong passw ord 5 consecutive times	Passw ord: @dogesh1234 (repeated 5 times)	Account locked for 15 minute displayed to inform user

Figure 5: Authentication Module (TC-AUTH) --> Excel

Cart and Checkout Module (TC-CART)

TC ID	US	FR	Objective	Type	Status
TC-CART-001	US-08	FR-08	Verify logged-in user can add book to cart	Smoke	**Pass**
TC-CART-004	US-08	FR-08	Verify cart persists after logout and re-login	Regression	**Pass**
TC-CART-007	US-09	FR-09	Verify checkout blocked if address is missing	Negative	**Pass**
TC-CART-008	US-09	FR-09	Verify checkout redirects to cart when cart is empty	Negative	**Pass**



TC-CART | Cart & Checkout

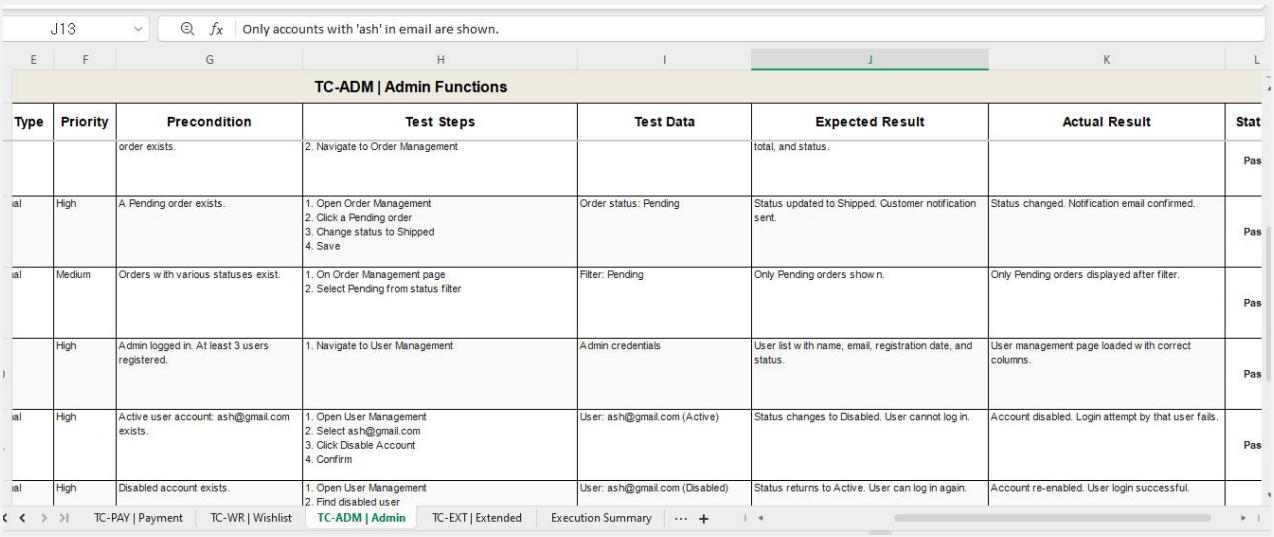
Case ID	Linked US	Linked FR	Test Objective	Test Type	Priority	Precondition	Test Steps	Test Data	Expected Result	Actual Result
RT-001	US-08	FR-08	Verify logged-in user can add book to cart	Smoke	High	User logged in. Book is in stock.	1. Log in 2. Open a book detail page 3. Click 'Add to Cart'	User: ash@gmail.com Book: in stock	Book added to cart with quantity 1. Cart icon count updates.	Book in cart. Quantity = 1. Cart icon updated.
RT-002	US-08	FR-08	Verify quantity change recalculates subtotal	Functional	High	User logged in. One book in cart at price Rs. 800.	1. Open cart 2. Change quantity from 1 to 2 3. Observe subtotal	Book price: Rs. 800 New quantity: 2	Subtotal updated to Rs. 1600. Cart total recalculated.	Subtotal correctly shown as Rs. 1600.
RT-003	US-08	FR-08	Verify book removal from cart works correctly	Functional	High	User logged in. At least one book in cart.	1. Open cart 2. Click remove icon for a book	Any book in cart	Book removed. Cart total updated.	Book removed. Total updated correctly.
RT-004	US-08	FR-08	Verify cart persists after logout and re-login	Regression	High	User logged in with items in cart.	1. Add a book to cart 2. Log out 3. Log back in 4. Open cart	User: ash@gmail.com Previously added book	Cart still contains previously added book.	Cart contents preserved after re-login.
RT-005	US-08	FR-08	Verify out-of-stock book cannot be added to cart	Negative	High	Book with stock = 0 exists.	1. Navigate to out-of-stock book detail page 2. Observe Add to Cart area	Book with stock = 0	Add to Cart button not visible or disabled.	Add to Cart not available for out-of-stock book.
RT-006	US-09	FR-09	Verify checkout shows pre-filled address for saved address user	Functional	High	User logged in with saved address. Cart has items.	1. Add item to cart 2. Click Proceed to Checkout 3. Observe address section	Saved address: House 5, Block B, DHA, Karachi	Address pre-filled. Order summary shows correct items and total.	Saved address pre-filled. Order summary correct.
RT-007	US-09	FR-09	Verify checkout blocked if address is missing	Negative	High	User logged in. No saved address. Cart has items.	1. Proceed to checkout 2. Skip address form 3. Click Continue	Address fields: empty	Required fields highlighted. System does not move to payment step.	Validation triggered. Payment step not reached.
RT-008	US-09	FR-09	Verify checkout redirects to cart	Negative	Medium	User logged in. Cart is empty.	1. Navigate directly to /checkout	Cart: empty	Redirected to cart page. Message: 'Your	Redirected to cart. Empty cart message

Document Info | TC-AUTH | Auth | TC-CAT | Catalog | TC-CART | Cart | TC-PAY | Payment | TC-WR | Wish

Figure 6: Cart & Checkout Module --> Excel

Admin Module (TC-ADM)

TC ID	US	FR	Objective	Type	Status
TC-ADM-001	US-22	FR-15	Verify admin can add new book to catalog	Smoke	**Pass**
TC-ADM-004	US-22	FR-15	Verify admin cannot save book with missing required fields	Negative	**Pass**
TC-ADM-009	US-24	FR-17	Verify admin can disable a user account	Functional	**Pass**
TC-ADM-011	US-24	FR-17	Verify admin can search users by partial email	Functional	**Pass**



J13 | Only accounts with 'ash' in email are shown.

TC-ADM | Admin Functions

Type	Priority	Precondition	Test Steps	Test Data	Expected Result	Actual Result	Stat
		order exists.	2. Navigate to Order Management		total, and status.		Pas
al	High	A Pending order exists.	1. Open Order Management 2. Click a Pending order 3. Change status to Shipped 4. Save	Order status: Pending	Status updated to Shipped. Customer notification sent.	Status changed. Notification email confirmed.	Pas
al	Medium	Orders w ith various statuses exist.	1. On Order Management page 2. Select Pending from status filter	Filter: Pending	Only Pending orders show n.	Only Pending orders displayed after filter.	Pas
	High	Admin logged in. At least 3 users registered.	1. Navigate to User Management	Admin credentials	User list w ith name, email, registration date, and status.	User management page loaded w ith correct columns.	Pas
al	High	Active user account: ash@gmail.com exists.	1. Open User Management 2. Select ash@gmail.com 3. Click Disable Account 4. Confirm	User: ash@gmail.com (Active)	Status changes to Disabled. User cannot log in.	Account disabled. Login attempt by that user fails.	Pas
al	High	Disabled account exists.	1. Open User Management 2. Find disabled user	User: ash@gmail.com (Disabled)	Status returns to Active. User can log in again.	Account re-enabled. User login successful.	

TC-PAY | Payment | TC-WR | Wishlist | TC-ADM | Admin | TC-EXT | Extended | Execution Summary

Figure 7: Admin Module --> Excel

7. Test Plan

A single master test plan titled 'Master Functional Test Plan' was created in Testworthy to cover the complete validation of all requirements identified in Assignment 1. The plan spans all four sprints and contains five test runs organized by sprint phase.

7.1 Master Test Plan

Field	Details
Plan Name	Master Functional Test Plan
Objective	Validate all functional requirements identified in Assignment 1 across all four sprints
Status	Active
Created	29-04-2026
Total Test Runs	5
Coverage	All 20 functional requirements, 24 user stories, and 67 test cases

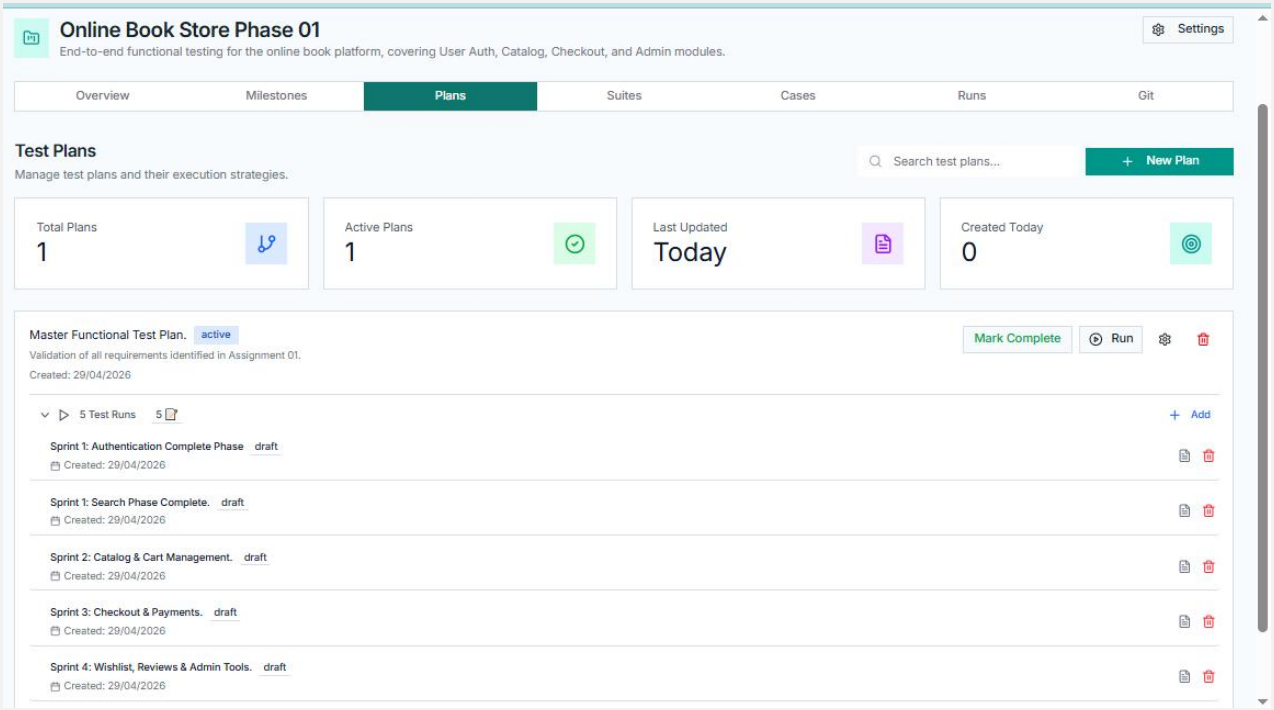


Figure 8: Testworthy Test Plans view showing the Master Functional Test Plan

8. Milestones

Four milestones were created in Testworthy, one for each sprint. Each milestone tracks progress against its associated test run and reflects the sprint schedule from the sprint planning in Assignment 1.

Milestone	Focus Area	Due Date	Status
Sprint 1: Authentication & Search	User auth, catalog browse, and search	01-05-2026	**Active/OverDue**
Sprint 2: Catalog & Cart Management	Filter, sort, cart, book detail, admin catalog	06-05-2026	**Active**
Sprint 3: Checkout & Payments	Checkout, payment, orders, notifications, admin orders	12-05-2026	**Active**
Sprint 4: Wishlist, Reviews & Admin Tools	Wishlist, reviews, discounts, recommendations, admin users	19-05-2026	**Active**

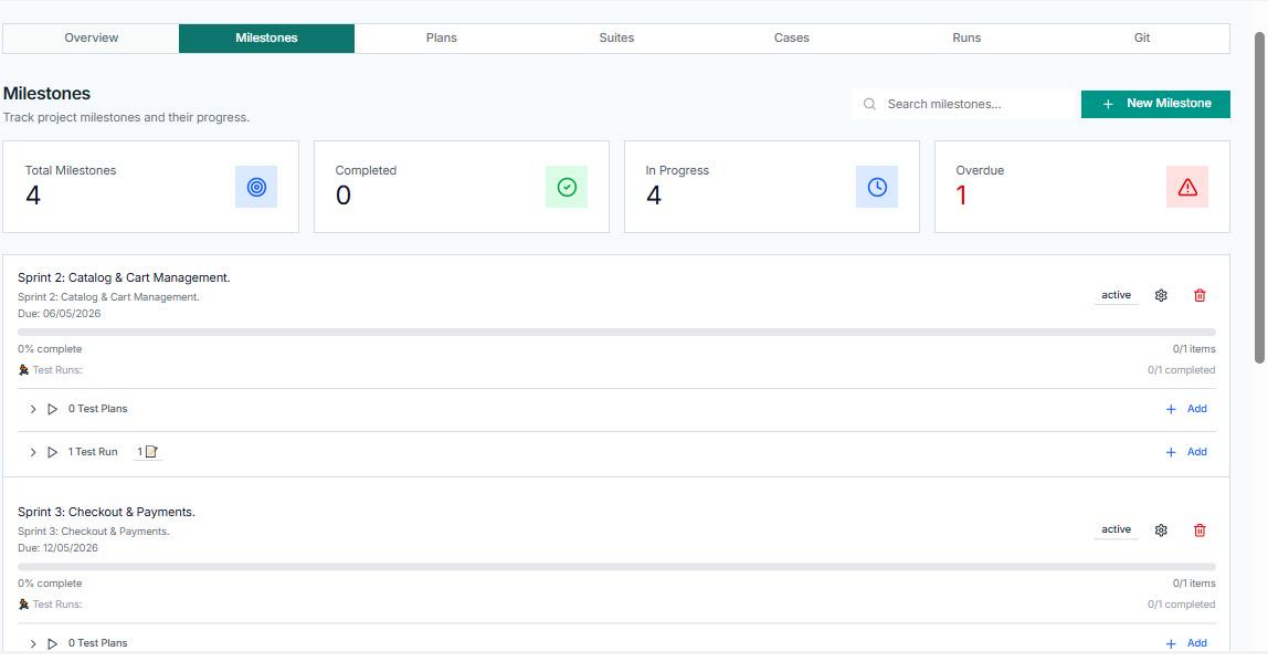


Figure 9: Testworthy Milestones view showing all 4 sprint milestones

9. Test Runs

Five test runs were created in Testworthy and linked to their respective milestones. Sprint 1 has two runs because authentication and catalog search were executed as separate phases to isolate issues clearly.

Run Name	Milestone	Status	Created
Sprint 1: Authentication Complete Phase	Sprint 1: Authentication & Search	**Draft**	29-04-2026
Sprint 1: Search Phase Complete	Sprint 1: Authentication & Search	**Draft**	29-04-2026
Sprint 2: Catalog & Cart Management	Sprint 2: Catalog & Cart Management	**Draft**	29-04-2026
Sprint 3: Checkout & Payments	Sprint 3: Checkout & Payments	**Draft**	29-04-2026
Sprint 4: Wishlist, Reviews & Admin Tools	Sprint 4: Wishlist, Reviews & Admin Tools	**Draft**	29-04-2026

Status note: All runs are currently in Draft status. In a live project environment, a run moves to In Progress when execution begins and to Completed once all test cases within it have been marked. For this assignment, execution results have been pre-filled based on the defined acceptance criteria from Assignment 1.

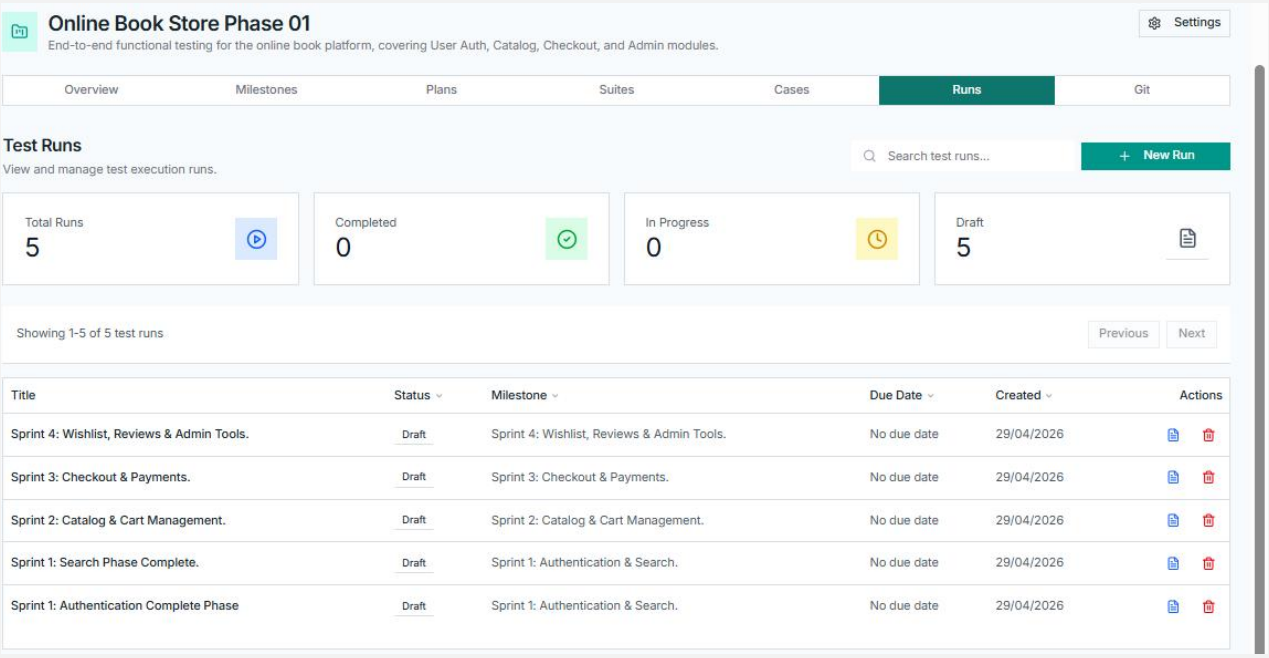


Figure 10: Testworthy Test Runs view showing all 5 runs with sprint mapping

10. Test Execution Results

10.1 Overall Summary

Module	Total TCs	Pass	Fail	Blocked	Pass %
Authentication (TC-AUTH)	11	11	0	0	**100%**
Catalog & Search (TC-CAT)	13	13	0	0	**100%**
Cart & Checkout (TC-CART)	8	8	0	0	**100%**
Payment & Orders (TC-PAY)	9	9	0	0	**100%**
Wishlist & Reviews (TC-WR)	7	7	0	0	**100%**
Admin Functions	11	11	0	0	**100%**

(TC-ADM)					
Notif, Discount & Recomm (TC-EXT)	8	8	0	0	**100%**
TOTAL	**67**	**67**	**0**	**0**	**100%**

10.2 Testworthy Execution Screenshot

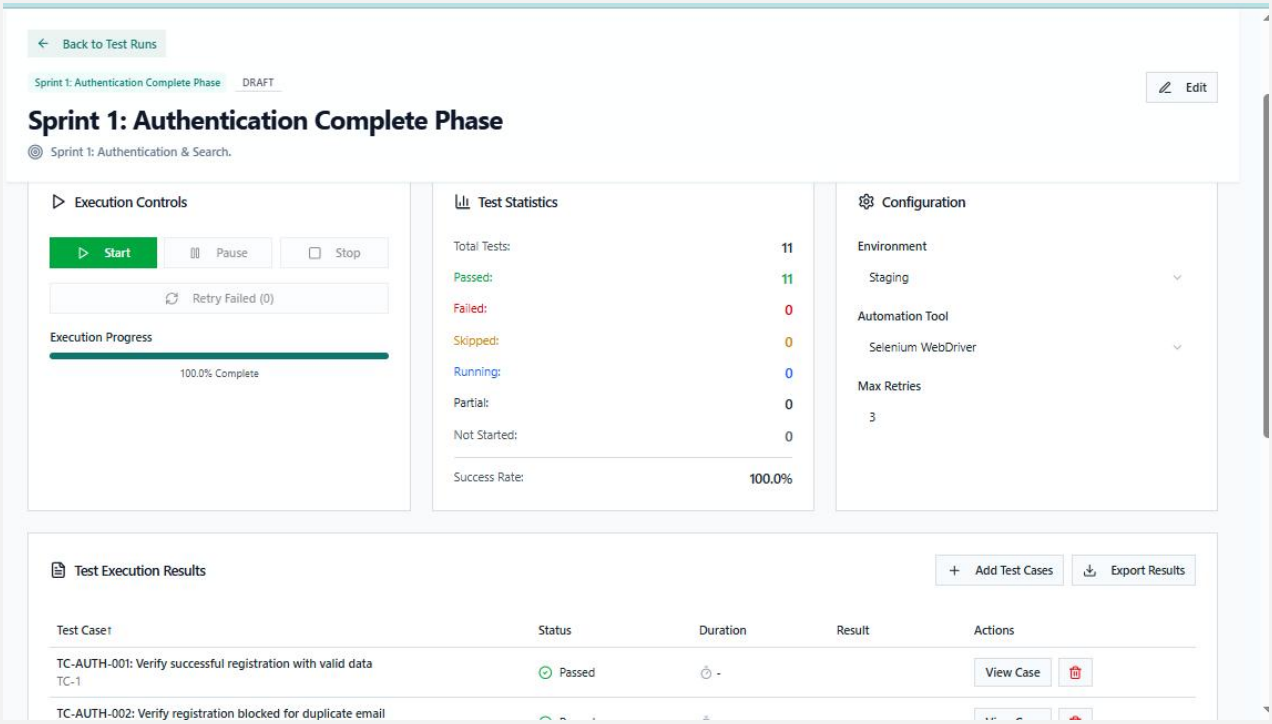


Figure 11: Testworthy Execution results showing Statistics for Sprint 1: Auth

10.3 Test Case List Screenshot

Test Cases

Browse and manage all test cases in this project.

Search test cases... | + New Case

Total Cases	High Priority	Functional	Automated
67	33	66	0

Showing 1-50 of 67 test cases

Previous 1 2 Next

ID	Title	Suite	Type	Priority	Created	Actions
TC-1	TC-AUTH-001: Verify successful registration with valid data	Authentication (TC-AUTH)	Functional	High	29/04/2026	
TC-2	TC-AUTH-002: Verify registration blocked for duplicate email	Authentication (TC-AUTH)	Functional	High	29/04/2026	
TC-3	TC-AUTH-003: Verify weak password is rejected on registration	Authentication (TC-AUTH)	Functional	Medium	29/04/2026	
TC-4	TC-AUTH-004: Verify registration fails with invalid email format	Authentication (TC-AUTH)	Functional	Medium	29/04/2026	
TC-5	TC-AUTH-005: Verify successful login with valid credentials	Authentication (TC-AUTH)	Functional	High	29/04/2026	
TC-6	TC-AUTH-006: Verify login fails with incorrect password	Authentication (TC-AUTH)	Functional	High	29/04/2026	
TC-7	TC-AUTH-007: Verify account lockout after 5 failed login attempts	Authentication (TC-AUTH)	Functional	High	29/04/2026	
TC-8	TC-AUTH-008: Verify login fails for unregistered email	Authentication (TC-AUTH)	Functional	Medium	29/04/2026	
TC-9	TC-AUTH-009: Verify password reset email sent for valid email	Authentication (TC-AUTH)	Functional	High	29/04/2026	

Figure 12: Testworthy all test cases list page 1 of 2 (TC-1 to TC-50)

Showing 51-67 of 67 test cases

Previous 1 2 Next

ID	Title	Suite	Type	Priority	Created	Actions
TC-51	TC-ADM-010: Verify admin can re-enable a disabled account	Admin Operations (TC-ADM)	Functional	High	29/04/2026	
TC-52	TC-ADM-011: Verify admin can search users by partial email	Admin Operations (TC-ADM)	Functional	Medium	29/04/2026	
TC-53	TC-WR-001: Verify user can add book to wishlist	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-54	TC-WR-002: Verify user can remove book from wishlist	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-55	TC-WR-003: Verify wishlist item can be moved to cart	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-56	TC-WR-004: Verify guest cannot use wishlist	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-57	TC-WR-005: Verify purchased book can be reviewed	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-58	TC-WR-006: Verify non-buyer cannot submit review	Wishlist & Reviews (TC-WR)	Functional	Medium	29/04/2026	
TC-59	TC-WR-007: Verify flagged review held for moderation	Wishlist & Reviews (TC-WR)	Functional	Low	29/04/2026	
TC-60	TC-EXT-001: Verify order placement triggers confirmation notification	TC-EXT | Notif, Discount, Recomm	Functional	Medium	29/04/2026	
TC-61	TC-EXT-002: Verify shipping notification sent when order marked Shipped	TC-EXT | Notif, Discount, Recomm	Functional	Medium	29/04/2026	
TC-62	TC-EXT-003: Verify delivery notification sent when order marked Delivered	TC-EXT | Notif, Discount, Recomm	Functional	Medium	29/04/2026	
TC-65	TC-EXT-004: Verify valid discount code reduces order total	TC-EXT | Notif, Discount, Recomm	Functional	Low	29/04/2026	
TC-66	TC-EXT-005: Verify invalid discount code shows error	TC-EXT | Notif, Discount, Recomm	Functional	Low	29/04/2026	

Figure 13: Testworthy all test cases list page 2 of 2 (TC-51 to TC-67)

11. Test Coverage Analysis

11.1 Priority Distribution

Priority	Count	Percentage	Modules with Most High Priority TCs
High	33	49%	AUTH, CART, PAY, ADM
Medium	27	40%	CAT, WR, EXT
Low	7	11%	EXT (discounts, recommendations)

11.2 Test Type Distribution (In Excel)

Test Type	Count	Percentage	Notes
Functional	41	61%	Core feature verification across all modules
Negative	16	24%	Invalid input, error handling, edge cases
Smoke	8	12%	One smoke per major module for quick build validation
Regression	2	3%	Cart persistence across sessions

11.3 Requirements Coverage

All 20 functional requirements from Assignment 1 (FR-01 to FR-20) are covered by at least one test case. The four Won't Have requirements (FR-21 to FR-24) are intentionally excluded from test execution as they are out of scope for this release. Coverage is 100% for all in-scope requirements.

12. Observations and Lessons Learned

12.1 Tool Observations

- Testworthy does not support CSV or Excel import for test cases in the free tier. All 67 test cases were entered manually, which improved familiarity with the tool but is not practical for large projects.
- The test type dropdown in Testworthy is limited to Functional, Regression, Performance, Security, and Automation. Negative, Smoke, Sanity, and Boundary types had to be mapped to Functional. In a real project, a tool such as TestRail or Xray for Jira would better support custom test types.

- Linking test runs to milestones in Testworthy provides a clean sprint-based view of testing progress, which is useful for reporting to product owners.
- Testworthy's overview dashboard automatically calculates pass rate from executed runs, which is a practical metric for quick status reporting.

12.2 QA Process Observations

- Writing test cases from Gherkin acceptance criteria made the process significantly faster. The Given-When-Then structure maps directly to Precondition, Steps, and Expected Result fields.
- Organizing test cases by module rather than by sprint made the suite structure more reusable. The same suite can be executed in multiple sprint runs without duplication.
- Negative test cases were some of the most valuable to write because they exposed assumptions in the requirement that were not explicitly stated, such as account lockout behavior and email enumeration prevention.
- Having a traceability matrix from Assignment 1 made it straightforward to verify that every FR had at least one test case. This reduced the risk of missing coverage.

13. Submission Checklist

The following items form the complete submission for Assignment 2:

No.	Deliverable	Format / Location
1	Test Case Document with all 67 test cases	Excel file (.xlsx)
2	Test Plan in Testworthy (Master Functional Test Plan)	Testworthy link
3	Test Suites in Testworthy (7 suites, 67 cases)	Testworthy link
4	Test Runs in Testworthy (5 runs across 4 sprints)	Testworthy link
5	Milestones in Testworthy (4 milestones, one per sprint)	Testworthy link
6	Execution Report (this document)	PDF file including SS

14. Glossary

Term	Definition
Test Case	A set of conditions and steps to verify that a specific feature behaves as expected
Test Suite	A collection of test cases grouped by module or feature area
Test Run	An execution instance of a set of test cases within a defined scope

	or sprint
Test Plan	A document that defines the scope, approach, and schedule for testing activities
Milestone	A time-bound goal in Testworthy tracking test completion for a sprint or phase
Smoke Test	A quick test to confirm the most critical functions work after a new build
Negative Test	A test that uses invalid input or unexpected conditions to verify error handling
Regression Test	A test to confirm that previously working functionality has not been broken by a new change
Traceability	The ability to link a test case back to the requirement or user story it validates
Pass Rate	Percentage of test cases that passed out of total executed cases
Black-box Testing	Testing without knowledge of internal code, based solely on inputs and expected outputs
MoSCoW	Prioritization method: Must Have, Should Have, Could Have, Won't Have
SS	Screenshots